

Exploring the synergy between generative AI and software engineering: Automating code optimization and bug fixing

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Abstract

As applied to software engineering, generative AI is quickly transitioning from a zero-sum industry game changer into the primary automation tool for code optimization, bug identification, and problem-solving. This technology takes advantage of artificial intelligence algorithms within machine learning models to analyze and write code, resulting in improved quality and speed of an application development process. The generative AI replenishes productivity in development work and enhances centralization between development work teams through code handling and intelligent suggestions for essential codes. However, the integration of AI in software engineering poses the following problems and ethical questions: the question of accuracy, bias, and data. This paper will review the existing knowledge on generative AI in software engineering regarding its current use, future evolutions and advancements, issues and limitations, and ethical factors in using this technology. This paper considers these aspects to give a global outlook on how generative AI will transform software development in the future and how responsible AI should be employed.

Keywords: Generative AI; Software engineering; Code optimization; Bug detection; Automated debugging; Developer productivity

1. Introduction

Generative AI is changing software engineering practices by introducing enhanced automation, effectiveness, and product quality. With the use of software in daily operations and functionalities in financial, healthcare, social media, and other sectors, the need for better, faster, efficient, and reliable methods to develop software is ever on the rise. The best conventional software engineering practices may include activities such as redundancy, test and retest, computer checks, debug which may take a lot of time and lead to delays in the engineering process and also increase cost.

The generation of AI based on generative data models promises to advance opportunities for optimizing different facets of software engineering. In contrast to AI, where humans define rules guiding the system and supervised learning, generative AI applies effective models, including deep learning neural networks, to learn human language patterns and generate human-like texts, codes, and other outputs. Based on vast data, such models can adopt the peculiarities of specific programming languages, architectures, and development paradigms. They can help developers by performing repetitive tasks such as code optimization and more accurate bug finding and sometimes provide solutions for these bugs, saving the developers considerable time.

This paper explores the transformative potential of generative AI in software engineering, focusing on three critical areas: code optimization, bug identification, and bug fixing. In this paper, by outlining the application of generative AI in the indicated domains, we will show how generative AI optimizes, secures, and improves software engineering. When hybrid generative AI tools are applied to the software development process, they not only speed up the coding process

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but also increase the code quality and enhance the capability that is difficult for humans to reach, which indicates the potential for development in the future of software engineering.

As we discuss these topics, we'll also discuss the difficulties and ethical concerns linked with generative Integrated AI to avoid presenting only positive aspects of this technology. Finally, this paper concludes that generative AI is still in its early stages of development, but it has the potential to be a game changer in software engineering and make the environment for developing software more effective and creative.

2. The evolution of software engineering and AI

Software engineering has grown measurably over the years from just pure coding or programming projects into larger system projects. In the beginning, software engineering had only simple things, such as coding, where developers wrote programs and debugged them by trial and error. With time and as the world's technology grew, the engineering field broadened to structured programming, object-oriented design, and Agile engineering. Nevertheless, there are or have been some challenges that have or can be linked to software engineering regardless of the advancements, such as;

Integrating artificial intelligence in software engineering was considered a revolution in developing that area. At first, AI was used as icing on the software development cake, where its use was restricted to certain practicalities, like code suggestion and predictive text, and it worked according to a set of rules and datasets. These early applications served their purpose but had many constraints and limitations when used in practical applications. While they didn't transform the software development process, they added some automation capabilities to existing traditional processes.

Generative AI, something relatively new in the AI world, is a new approach to the application of AI in the SE. Unlike earlier-generation AI technologies, generative AI utilizes deep neural networks and sophisticated machine learning algorithms to comprehend and produce elaborated data outputs like code. These models are fed with large datasets, covering a broad spectrum of programming languages, frameworks, or coding standards, and, therefore, are capable of pattern recognition beyond the capability of the basic reinforcing AI. Thus, an AI falls under generative AI, which can be applied to tasks such as writing code, editing existing code bases, and even putting forward changes in the overall architecture of the program – rather than simply being limited to repetitive tasks.

The most crucial advantage of generative AI in software engineering is code optimization. When examining a sample code, generative AI can distinguish areas that deserve optimization to increase the code's efficiency and stability. This level of automated code optimization was impossible before, and even now, it is a giant leap forward compared to the previous approach, when each code must be rewritten with minimum effort by the expert.

A second and pivotal use case of generative AI is to upgrade bug discovery. This means that it becomes crucial to underscore the fact that traditional bug detection techniques entail the use of tools like static code analyzers, source code audits, and manual code review; this method is relatively very inefficient and can easily miss the bugs or/and vulnerabilities in the large complex software systems. In addition, Generative AI applies the learning from the approach where the device can consider patterns connected to usual vices and imperfections, which makes it superior to Discriminative AI in detecting defects in the early stages of software development.

Furthermore, generative AI can optimize the fix process to some extent. Now, with the help of AI, it is possible not only to 'point-and-click' a bug but also to understand why this bug occurred and, in some cases, suggest/ implement the solution to this problem. This capability not only helps to reduce the time taken to debug a program, but it also optimizes the efficiency of the solution given because, for instance, AI can use a lot of sample sizes of bugs and solutions to provide the best solutions.

Before delving deeper into the opportunities available with the help of generative AI in software engineering, we need to consider the drawbacks and problems of this technology. When adopting AI in development, some of the issues must be addressed: accuracy, compatibility of the models with the development tools used, data privacy and security, and bias in the generated code. However, there are several challenges that may come with generative AI in software engineering, and even with many AI and machine learning systems, the future that it brings is bright, full of features, fun, and efficiency in the production of software in a very short period compared to the previous years.

3. Code optimization with generative AI

Optimization is an elementary process of software engineering that mainly concerns the performance enhancement of designs and their subsequent codes. In simple terms, code optimization has been more of a practice whereby developers read through code and look for issues that could slow down the application before rewriting it in an improved form. This process can be time-consuming but can be performed very effectively only when one properly understands the code and the domain it serves. The emergence of generative AI changes the way code optimization can be achieved significantly, and new prospects for its automation open up in this direction.

True generative AI can quickly go through extensive and complicated codebases and determine in which areas performance may be enhanced or in which coding paradigms could be optimized. Totalitarian tools differ from rule-based tools, which allow for analyzing only simple code fragments or pointing at obvious syntactical errors or obvious inefficiencies of code fragments, in that AI models can learn from the examples and identify more sophisticated patterns of code behavior. These models have excellent database feeds that consist of well-optimized code that is used to train them to identify good practices and suggest further optimizations that are good practices.

Furthermore, generative AI can help decide the best place for the current code to improve performance and recognize what part of it is most resource-consuming. This process is critical today when resource limitations and response times are essential, for example, in real-time systems, applications used in high-frequency trading, or mobile devices. AI models can organize runtime behavior and determine program bottlenecks before the latter's manifestation. This helps optimize—for instance, breaking down some executions to work on multiple cores or providing other algorithms faster than others for data manipulation. This can enhance execution time and resource allocation, thus providing our customers with a better experience while lowering our operational overheads.

More concrete usages of generative AI applied to code optimization are gaining popularity across different fields today. For instance, high-performance software firms like gaming studios and the financial service sectors have embraced AI tools to automate the code. AI tools applied to coding have tested capabilities of trimming the size of codes and enhancing other measures of performance, including the rate of processing and amount of memory used. When comparing the old and new code repositories optimized using AI, you can always see a significant jump, which shows how useful AI is in this context.

Nevertheless, as explained before, applying generative AI to optimize code is challenging. A significant issue is the applicability of optimized solutions developed through AI algorithms. Even though an AI model can process considerable amounts of data, it can make mistakes and only sometimes propose a solution that is ideal for a project given a set of very stringent rules and regulations. Hence, it is essential for the developers who integrate AI solutions to check the output for quality constantly and verify whether the optimizations it has drawn impose different problems.

Finally, generative AI is reshaping the world of code optimization in software engineering. Automating refactoring makes tasks much faster and easier or way more efficient and ensures that memory usage needs to be increased as well so that developers can spend their time better dealing with these tasks. AI technology is expected to play an increasingly important role in code optimization and can be considered an essential tool in a software engineer's tool chest.

Table 1 AI Tools for Code Optimization and Bug Fixing

Tool Name	Functionality	Techniques Used	Strengths	Limitations	Examples of Use Cases
DeepCode	Bug detection, Code review	Machine Learning, Static Analysis	High accuracy in detecting bugs and code smells	Limited language support, primarily focused on JavaScript	Used by companies to automate code review processes
Codex	Code generation, Bug fixing	NLP, Transformer Models	Ability to generate code snippets and fix common bugs	May produce incorrect or suboptimal code	Integrated into GitHub Copilot for assisting in coding
Snyk	Security vulnerability	Machine Learning, Static Analysis	Focus on identifying and	Primarily focused on security, not general bug fixing	Used by organizations for

	detection, Bug fixing		fixing security vulnerabilities		enhancing security in codebases
SonarQube	Code quality analysis, Bug detection	Static Analysis, Rule-based Engine	Comprehensive code quality and security checks	Requires configuration, may produce false positives	Widely used for continuous /integration continuous delivery (CI/CD) pipelines
TabNine	Code completion, Bug fixing	Transformer Models, Deep Learning	Efficient code completion and bug suggestions in real-time	Primarily assists in code completion, not deep bug fixing	Used in IDEs for enhancing developer productivity
Linting Tools (e.g., ESLint, Pylint)	Code quality checks, Bug detection	Rule-based Static Analysis	Highly customizable, real-time feedback	Limited to rule-based checks, no learning capabilities	Used to enforce coding standards and catch basic bugs early in the development process
Infer	Bug detection, Code analysis	Static Analysis, Formal Verification	Strong in detecting null pointer exceptions, race conditions	Focuses on specific bug types, may miss broader issues	Used by large tech companies to enhance code reliability

4. Bug detection through generative AI

Testing for bugs is an essential phase of software development since it helps ascertain the appropriateness and effectiveness of code. Otherwise, bug detection was a more primitive and time-consuming process, relying on code reviews, static analysis tools, and rigorous testing to locate errors and flaws in the code. These methods are helpful, although time-consuming, and may not pick complicated or concealed problems. The application of generative AI to bug detection is a breakthrough, as it provides a more complex and automatic search for bugs and vulnerabilities in the code.

More specifically, generative AI uses deep learning models trained on tons of code and bug reports to identify patterns attributable to errors and, in general, security threats. Compared with static analysis tools filled with specific patterns and rules, generative AI models can comprehend the code's context and intentions. This capability enables AI DL-based testing to catch simple errors, which are simple to identify from the surface, and complicated issues, which may be due to the interaction between various parts of the software.

Among the key benefits of adopting generative AI for bug detection, the static code analysis advantage could be enumerated on top of the others. Static code analysis tools of the past provide extensive data that is frequently false and can be misleading to developers as real problems may get lost in the sea of false positives. While generative models are more accessible to create and more likely to generate false positives, non-generative models can lower false favorable rates over time and discern between legitimate problems and standard code. It helps AI learn through these experiences, providing them with better accuracy and solutions to real problems, for which the concentration of developers needs to be given.

Also, generative AI can be applied to such important tasks as identifying security threats in software code since the number of cyber threats tends to grow. Machine learning algorithms can be pre-taught to detect patterns linked with typical security risks. Thus, they offer a more preventive security approach, which may deal with threats before they are exploited. In addition, this capability enhances software security and minimizes the time-consuming and resource-draining of other types of security reviews.

Generative AI detecting bugs and potential issues has quickly evolved into an applicable and efficient tool in several real-life settings. Software companies developing sophisticated software systems in their portfolios from the finance and healthcare industry, among others, have adopted the use of AI-driven bug detection tools. All these tools have proven to have the potential to detect other bugs not spotted by manual testing, hence producing better, sometimes more secure software. For example, by utilizing AI models in applications, it is possible to find weak points in web

applications that have not been detected by traditional static analysis tools, which leads to improving the security of applications and preventing possible exploits.

However, there are some issues people have to face while using generative AI for bug identification. First, an organization's common challenge is the lack of quality training data to feed into the AI models. The datasets, as with any other data, may contain bias and outliers, which may, in turn, mean that the AI models may miss some bugs, especially those types of bugs that can only be detected with an exhaustive search strategy or, on the other end, create false alarms. Also, developers integrate AI into the existing developmental framework, making it sensitive to other developmental tools. It is also essential for developers to understand potential constraints involving AI and should always cross-check any results that AI generates.

There is a change in the detection of bugs in software development through generative AI. AI can help detect more bugs and vulnerabilities than the regular approach since AI uses more sophisticated pattern matching and anomaly detection algorithms. On the same note, it is also essential to look at the opportunities provided by AI for bug detection and exclusion. As far as the opportunities provided by AI in bug detection and exclusion are concerned, the potential is relatively large. With AI technology advancement, ensuring the quality and security of the software will become more crucial in the future, which will significantly contribute to and provide developers with new tools to improve their operation and development environments for improved software outcomes.

5. Bug resolution with generative AI

Bug fixing is another phase in the software development life cycle, which includes identifying, analyzing, and fixing the defects. Pre-ITIL, this process has been largely manual, where a developer has to determine the root cause of a bug, create a solution, and then test to confirm that it did not bring in other problems. Conversely, static binaries compiled from such code can take time to analyze and may be more strain-filled, especially in large or old projects. The emergence of generative AI is disrupting this process because it introduces automation into the process of bug resolution and thus saves a lot of time for developers to solve problems with software defects.

With generative AI, bug remediation is mainly achieved through auto-code correction, which enhances the process of identifying bugs' root causes much faster. Effort in debugging usually has been a lengthy and very arduous process that entails going through the codes by hand and using other debugging resources to find the origin of the error. On the other hand, generative AI models can process large codebases within the shortest time possible and give information on where and how the bug has happened. With the help of machine learning techniques, these identified models can understand in which context the particular bug appears and in which function or module and find the potential causes of the bug by learning from similar data patterns as experienced before.

AI does not only help shorten the time it takes to debug but also makes it more precise than before. However, Generative AI can recognize intricate problems that may go unnoticed by human developers during the coding process; these can be logical errors, concurrent issues, or any other case that is difficult to reproduce during testing. For instance, ML can simulate the various control flows throughout the program and give the developer the exact environment under which a race condition could occur. This level of detail, of course, can be beneficial in various complex bugs that, otherwise, will take a lot of time and effort and unique skills to be discovered.

Including the patches in the CI/CD cycle makes the bug resolution even more accessible for developers. CI/CD is a Continuous Integration and Continuous Deployment process where code changes created are checked and launched through the various stages to ensure that new bugs installed by recent changes are quickly detected. Introducing generative AI into this process means that organizations cannot only automate bug identification but also bug fixing, thus leading to faster time-to-deployment of crucial patches. This capability proves to be helpful in contexts where there is a need to start and deliver many cycles consecutively within a short span of time, for instance, web developments and cached agile software projects.

Several uses of generative AI in bug resolution show the impact and benefits it can bring towards enhancing and optimizing the quality of developed software while minimizing cost. For instance, firms in high-risk sectors such as finance or healthcare that need software to be highly reliable have started employing AI algorithms to develop patches for all the known types of vulnerabilities. These tools have been shown to help reduce the number of open bugs and the time taken before a software product can become stable to be released to the market. Also, AI-integrated bug resolution has been found to make resolutions more dependable and uniform because the models work from an extensive database, hence applying the correct bug-fixing methods.

However, the application of generative AI in bug resolution has some disadvantages as follows. One issue is the sustainability of the AI-created patches. While it makes changes based on the learned patterns, it needs to fully understand the business rules or even the specifics of every application. Consequently, modifying code by an algorithm – be it through machine learning or rule-based reasoning – might fix the problem at hand while creating a host of other unforeseen problems or be negligent of the consequences throughout the rest of the system. Hence, it becomes crucial to study and validate the patches developed by the AI system before implementing them on live platforms.

Another issue is guaranteeing the quality and relevance of datasets to AI models deployed for bug resolution. If the training data is inadequate or the data collected is biased, it will affect the AI's patch generation capability. For the models to be continuously improved, new data has to be introduced frequently, together with feedback from developers who rely on the solutions implemented by the AI systems.

Generative AI will significantly impact bug resolution in software engineering. Debugging and patch generation with the help of AI makes the bug-fixing process less time-consuming and brings out signaling improvement in software development. Despite the prospects of AI in bug resolution, there are still some issues, such as the validity of the patches that the AI has developed, but the appeal of using AI to solve bugs must be considered. The requirement of integrating AI technologies into software development will continue to amplify as the technology keeps evolving in the future, making bug handling and resolution more efficient than ever to deliver more effective and efficient software systems.

6. Enhancing developer productivity and collaboration

Modern generative AI is changing concrete technological features of software development like code refactoring and debugging and constantly increasing developers' overall individual and team efficiency. When adopting AI tools in the software development processes, there comes a benefit of automated development, thus reducing the manual, repetitive tasks that are time-wasting and improving teamwork in the development processes to be effective and efficient.

The primary benefit of generative AI to developers is the improvement of productivity due to the automation of numerous monotonous work. Some tasks like code review, documentation, or unit testing often consume a lot of time for developers who do not deal with creative or analytical thinking. Generative AI can further handle these tasks by generating code suggestions, writing documentation inspired by code analysis, and developing test cases. For instance, code completion apps powered by artificial intelligence, such as GitHub Copilot, generate code as developers type, suggesting different options that developers might want to implement. Such automation is helpful as it allows developers to spend more time on time-consuming tasks such as problem-solving or creative processes rather than on routine details, making them more effective and satisfied with their work.

It also helps enhance the quality of the codes and their maintainability, which, in any case, directly affects productivity. In this case, AI programs give real-time feedback on coding practices, which assists developers in practicing what is considered suitable in coding and avoiding what is wrong, which may lead to bugs or technical debts. This makes the effort to proactively ensure quality throughout the development process much less time-consuming than adding quality to the code later when it is already written. Over time, this can result in more stable and reliable software and a reduction in the cost of software development.

They also help collaborate by improving communication amongst development teams in generative AI. For example, through self-writing documents generated based on code repositories, documentation can be easily created or updated to enhance comprehensiveness and reduce the mental effort required by developers as they review or work on another programmer's code. This is especially useful when working with many people, especially when developers only see each other sometimes. High-quality and well-structured documentation created by AI helps minimize the communication gap between the team members. Therefore, the team is aligned concerning the goals of the project, its structure, and its code.

In different sectors, the application of generative AI to increase productivity and cooperation between developers is well-advanced. Large-scale technological organizations and businesses that develop complex software or dispersed teams are inclined to apply AI to optimize development procedures and team collaboration. For instance, Microsoft and Google have applied artificial intelligence-driven integrated development environments that allow code suggestions, testing, and continuous integration to increase developers' productivity.

However, several drawbacks and risk factors are evident when applying generative AI to productivity and improved teamwork. Another one is that people who solve problems using AI tools will reduce the exercise of developers' capabilities and their inherent critical thinking abilities. When it comes to development, different layers of development

teams should also find the right middle and make sure that AI is used to make work more accessible while keeping the creative processes and logical decision-making in check. On the same note, data privacy and security matters spring up, especially when using artificial intelligence and related applications that need extensive databases to use codes belonging to a particular company or entity. It is also imperative that the identified tools are developed in a way that would comply with the regulations of the respective industries.

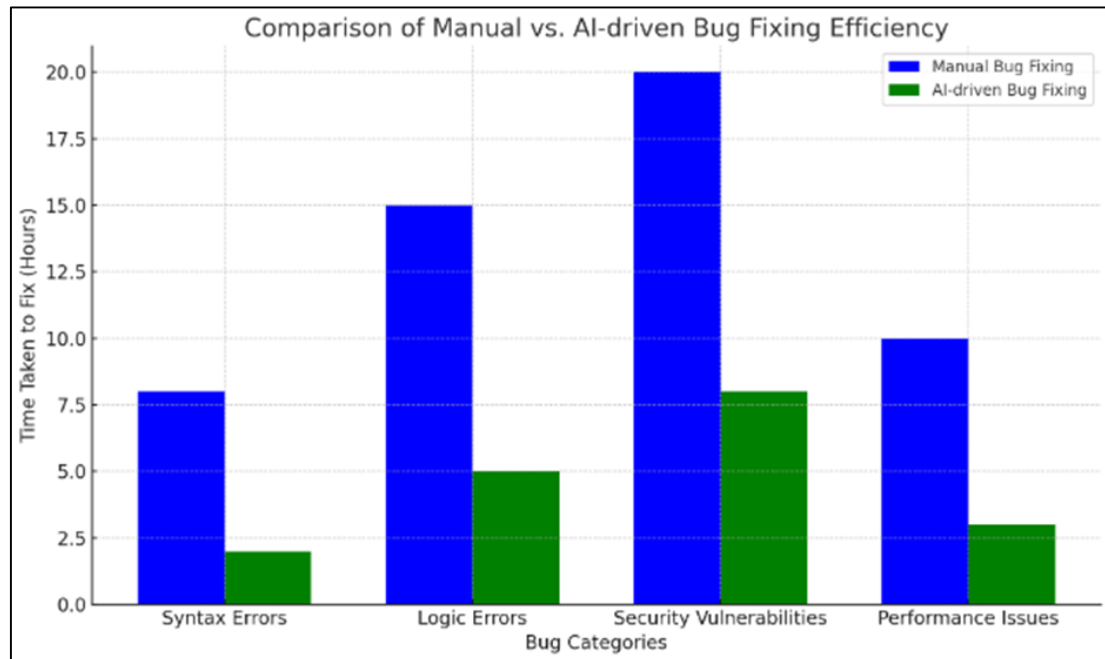


Figure 1 Comparison of Manual vs. AI-driven Bug Fixing Efficiency

7. Challenges and ethical considerations

Generative AI has several advantages in SE, such as code optimization, bug identification, and improved productivity. However, its implementation in development cycles is only possible with possible issues and ethical concerns. Solving these problems is imperative so that AI tools can be used appropriately.

One of the significant issues that generative AI poses in software engineering is the quality of the result that is going to be generated by AI models. It has been evident in all AI models, especially machine learning, that their performances can be attributed to the amount and quality of data they use to learn from. Thus, if the training data is skewed, lacking in some data points, or becomes old, then the suggestions or decisions provided by the AI shall also be deficient. This can result in syntactical mistakes in coding, a loophole in the security measures adopted, or an inefficient algorithm, which affects the quality of the software. Using AI-generated outputs is helpful but should be used with prudence and run through a rigorous code check or considered a suggestion from the AI.

Data privacy and security are other relevant ethical issues that must be addressed when employing generative AI in software engineering. AI tools depend on a firm's codebase, which may contain valuable information that a firm does not wish to disclose to others. It is also imperative to guarantee that AI tools will remain safeguarded alongside user privacy. This includes the meaning of data security, for instance, by encrypting data or controlling access to it and choosing AI-based tools that meet existing data protection standards. However, there is a concern that AI models may 'accidentally' train on sensitive and sometimes private data and reproduce it in a manner that may be detrimental to privacy and violate intellectual property rights. To address this risk, organizations must review the data that feed AI models and ensure proper security measures that would bar anyone from using such information in wrongdoings.

Scalability and Explainability are also significant ethical factors that must be considered paramount when applying generative AI in the operation of software developments. Based on deep learning models, the decision-making process can only be seen as a 'black box' as, even though models deliver their outputs, they cannot comprehensively explain how the decision was made. This lack of transparency can be disadvantageous in software engineering because in the process of code solving when coming up with a recommendation or fixing some anomaly, it is essential to be informed about

why a specific recommendation or change was made. Developments' AI tools have to be dependable, which signifies that they should be able to explain why they have provided such or such a result. Further development of the AI models that generate more Explainability is still under development.

Finally, there are ethical issues that relate to accountability or responsibility. In case the code is produced by an AI tool or where the code is used to fix bugs, difficulties arise in identifying who is at fault now and then, if there is any at all. Supposing the code generated using AI has a security weakness, or if it transforms into an issue or fails to complete a task correctly, who is the culprit, the implementing developer or the AI tool creator? When AI is implemented within the software development process, there should be better policies and statements to ensure a policy statement that does not give room for a lapse in accountability that comes with the/framework's role.

8. Future trends and innovations

Therefore, it is expected that generative AI will become a key player in software engineering and other fields and facilitate new trends and innovative movements in the future. The future of generative AI in SW will immediately improve not only the technical part of development but also teamwork and, in general, creativity.

One clear trend for the future is the emergence of better models that can interpret and write code with better context and accuracy. There are discriminating current AI models, such as those of large language models, where syntax studying has improved and code snippet generation. Nevertheless, future AI models will be enhanced, where the AI systems will be predefined to understand the project's complexity, the architecture of software, and the business logic. Such models will make better and more relevant codes with fewer modifications, facilitating faster coding.

This is compounded by the emergence of autonomous coding through artificial intelligence. Autonomous coding is the capability of an AI system where programmers or other stakeholders deliver just a broad set of guidelines and leave the rest of the coding process to the AI system. Such an approach could significantly decrease the time and expenses needed for developing such products and bring increased adaptability and flexibility into the development process, enabling firms to experiment with new concepts and continually refine them. It also could liberate software development from the past habits of coding, as new systems would allow ordinary people without a coding background to create programs using AI natural language inputs or elements on a screen.

Last, the application of generative AI in DevOps is expected to emerge more frequently. DevOps is the methodology that focuses on the integration and deployment of the software. Thus, AI can automate these processes, including testing, deployment, and monitoring. Some possible future characteristics are that new artificial intelligence models could prevent incidents in advance, decide the right deployment strategy based on real-time data feeding, and scale resources according to dynamic needs. Such a level of automation and intelligence would allow for more robust and self-improving software systems, which could proactively adapt to new problems and possibilities.

9. Conclusion

The attempts to incorporate generative AI into software engineering can be viewed as the paradigmatic shift in how coders think about the objects of their activity, as well as in what regard they strategize and optimize it. The promises of generative AI are not only the productivity gain as mundane tasks are automated, code quality is improved, and knowledge work becomes more sociable. It also allows developers to leverage it to think outside the box about where to focus their time. The enhancements of the utilization of artificial intelligence in code beautification, fault identification, and their rectification in computer programming have started revealing an enhanced mode of software development precision and productivity.

On the horizon for generative applications of AI, software engineering shows remarkable growth in potential for the future. As a result of breakthroughs in recent AI technologies, flexibility related to integration with new technologies, and self-programming codes, the evolution of software construction is already imminent. It stated that AI is going to improve collaborative processes, tailor developer education, and promote code ethicality, thereby making the entire software development process more flexible and adaptable.

However, there are specific issues or concerns regarding generative AI, which are legal or ethical issues that must be solved or met. It is imperative to point out the AI outputs' credibility, the successful incorporation of AI into the current processes, and the challenges of handling data privacy, bias, and transparency to fully enjoy AI's opportunities without

falling prey to its risks. Furthermore, the effects of AI on the developers and the responsibility gaps have to be filled in a positive approach when it comes to AI.

Machine learning has been recognized as a game-changer in software development, which can increase efficiency, improve code quality, and promote project cooperation. This technology is rapidly evolving and will gradually take the focus of many organizations and startups within the SDLC, continuously improving their efficiency and introducing new ideas. Therefore, by dealing with the questions and concerns in this chapter duly and minimizing potential adversarial effects as much as possible, the software engineering community can optimize generative AI's utility for creating better, fairer, and more significant software systems.

Compliance with ethical standards

Disclosure of conflict of interest

No conflict of interest to be disclosed.

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